LECTURE SHEET

EXERCISE-I

(Algebra of limits L.H.L., R.H.L. trigonometric limits limits tend to ∞)

LEVEL-I (MAIN)

Single answer type questions

Evaluation of Algebra of Limits

1. If $\underset{x\to 0}{Lt}\dfrac{((a-n)nx-\tan x)\sin x}{x^2}=0$, where n is non zero real number, then 'a' is

1) 0 2) $\dfrac{n+1}{n}$ 3) n 4) $n+\dfrac{1}{n}$

2. $\lim\limits_{x\to 0^+}\{ln(x^7+2x^6e+x^5e^2)-ln(x^2\sin^3 x)\}$ is equal to

1) 0 2) 1 3) 2 4) 3

3. Evaluate $\lim\limits_{x\to a}\dfrac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}},(a\neq 0)$.

1) $\dfrac{2}{3\sqrt{3}}$ 2) $\dfrac{4}{3\sqrt{3}}$ 3) $\dfrac{5}{3\sqrt{3}}$ 4) $\dfrac{7}{3\sqrt{3}}$

4. Let $f:R\to[0,\infty)$ be such that $\lim\limits_{x\to 5}f(x)$ exists and $\lim\limits_{x\to 5}\dfrac{(f(x))^2-9}{\sqrt{|x-5|}}=0$

1) 0 2) 1 3) 2 4) 3

5. If $f(x)=3x^{10}-7x^8+5x^6-21x^3+3x^2-7$, then $\lim\limits_{\alpha\to 0}\dfrac{f(1-\alpha)-f(1)}{\alpha^3+3\alpha}$ is

1) $-\dfrac{53}{3}$ 2) $\dfrac{53}{3}$ 3) $-\dfrac{55}{3}$ 4) $\dfrac{55}{3}$

6. If $\lim\limits_{x\to 2}\dfrac{\tan(x-2)\{x^2+(k-2)x-2k\}}{x^2-4x+4}=5$, then k is equal to:

1) 0 2) 1 3) 2 4) 3

7. $\lim\limits_{x\to 0}\left[\min(y^2-4y+11).\dfrac{\sin x}{x}\right]$ (where [.] denotes the greatest integer function) is

1) 5 2) 6 3) 7 4) Does not exist

8. The value of $\lim\limits_{x\to 0}\left(\left[\dfrac{100x}{\sin x}\right]+\left[\dfrac{99\sin x}{x}\right]\right)$ (where [.] represents the greatest integral function) is

1) 199 2) 198 3) 0 4) 197

9. $\lim\limits_{x\to 0}\dfrac{\sqrt{1-\cos 2x}}{\sqrt{2}x}$ is equal to

1) 1 2) –1 3) zero 4) does not exist

10. $\lim_{x\to 2}\left(\dfrac{\sqrt{1-\cos\{2(x-2)\}}}{x-2}\right)$

1) Equals $\dfrac{1}{\sqrt{2}}$ 2) Does not exist 3) Equals $\sqrt{2}$ 4) Equals $-\sqrt{2}$

11. If $f(x)=\begin{cases}\dfrac{\sin[x]}{[x]}, & \text{for } [x]\neq 0\\ 0, & \text{for } [x]=0\end{cases}$, where $[x]$ denotes the greatest integer less than or equal to x, then $\lim_{x\to 0} f(x)$ is

1) 1 2) 0 3) -1 4) does not exist

Evaluation of trigonometric Limits

12. If $\lim_{x\to 0}\dfrac{f(x)}{x^2}=a$ *and* $\lim_{x\to 0}\dfrac{f(1-\cos x)}{g(x)\sin^2 x}=b$ (*where* $b\neq 0$), then $\lim_{x\to 0}\dfrac{g(1-\cos 2x)}{x^4}$ is

1) $\dfrac{4a}{b}$ 2) $\dfrac{a}{4b}$ 3) $\dfrac{a}{b}$ 4) $\dfrac{b}{a}$

13. $\lim_{x\to\frac{\pi}{2}}\dfrac{(1-\sin x)(8x^3-\pi^3)\cos x}{(\pi-2x)^4}$

1) $-\dfrac{\pi^2}{16}$ 2) $\dfrac{3\pi^2}{16}$ 3) $\dfrac{\pi^2}{16}$ 4) $-\dfrac{3\pi^2}{16}$

14. $\underset{x\to\frac{\pi}{4}}{Lt}\dfrac{\sqrt{1-\sqrt{\sin 2x}}}{\pi-4x}=$

1) 2 2) -2 3) 0 4) Does not exist

15. $\lim_{x\to 1}\dfrac{1+\sin\pi\left(\dfrac{3x}{1+x^2}\right)}{1+\cos\pi x}$ is equal to

1) 0 2) 1 3) 2 4) 4

16. The value of $\lim_{x\to 1^-}\dfrac{1-\sqrt{x}}{(\cos^{-1}x)^2}$ is

1) 4 2) 1/2 3) 2 4) 1/4

17. $\lim_{x\to 0}\dfrac{(1-\cos 2x)(3+\cos x)}{x\tan 4x}$ is equal to

1) $-\dfrac{1}{4}$ 2) $\dfrac{1}{2}$ 3) 1 4) 2

18. $\lim_{x\to 0}\dfrac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is equal to

1) 2 2) -2 3) 1/2 4) $-1/2$

19. The integer n for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite nonzero number is

1) 1 2) 2 3) 3 4) 4

20. $\lim_{x\to\frac{\pi}{2}}\frac{\left[1-\tan\left(\frac{x}{2}\right)\right][1-\sin x]}{\left[1+\tan\left(\frac{x}{2}\right)\right][\pi-2x]^3}$ is

1) ∞ 2) $\frac{1}{8}$ 3) 0 4) $\frac{1}{32}$

21. $\lim_{x\to 0}\left(\frac{x-\sin x}{x}\right)\sin\left(\frac{1}{x}\right)$

1) equals 1 2) equals 0 3) does not exist 4) equals -1

22. The value of $\lim_{x\to 0}\frac{1}{x}\left[\tan^{-1}\left(\frac{x+1}{2x+1}\right)-\frac{\pi}{4}\right]$ is:

1) 1 2) $-\frac{1}{2}$ 3) 2 4) 0

23. If $\lim_{x\to 0}\frac{ae^x-b}{x}=2$ then $a=$

1) 7 2) 4 3) 2 4) 1

24. $\lim_{h\to 0}\frac{\ln(1+2h)-2\ln(1+h)}{h^2}=$ ______

1) -1 2) 1 3) 2 4) 3

Limits of the form tends to infinite

25. $\lim_{n\to\infty}\sum_{r=1}^{n}\cot^{-1}(r^2+r+1)=$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) 0

26. $\underset{n\to\infty}{Lt}\, n^2\left(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}\right), x>0$, is equal to

1) 0 2) e^x 3) $\log_e x$ 4) 1

27. $\lim_{n\to\infty}\sum_{x=1}^{20}\cos^{2n}(x-10)=$

1) 0 2) 1 3) 20 4) 10

28. $\underset{n\to\infty}{Lt}\sum_{r=1}^{n}\cot^{-1}(2r^2)=$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) 0 4) π

29. If a_n and b_n are positive integers and $a_n+\sqrt{2}b_n=\left(2+\sqrt{2}\right)^n$, then $\lim_{n\to\infty}\left(\frac{a_n}{b_n}\right)=$

1) 2   2) $\sqrt{2}$ 3) $e^{\sqrt{2}}$ 4) e^2

30. If $\lim_{n\to\infty}\frac{n3^n}{n(x-2)^n+n.3^{n+1}-3^n}=\frac{1}{3}$, then the range of x is (where $n\in N$)

1) $[2,5)$ 2) $(1,5)$ 3) $(-1,5)$ 4) $(-\infty,\infty)$

31. The value of the limit $\lim_{x\to 0}\frac{a^{\sqrt{x}}-a^{1/\sqrt{x}}}{a^{\sqrt{x}}+a^{1/\sqrt{x}}}, a>1$, is

1) 4   2) 2   3) -1 4) 0

32. $\lim_{x\to\infty}\{(x+5)\tan^{-1}(x+5)-(x+1)\tan^{-1}(x+1)\}$ is equal to

1) π 2) 2π 3) $\pi/2$ 4) $3\pi/2$

33. $\lim_{n\to\infty}\frac{1+2^4+3^4+.....+n^4}{n^5}-\lim_{n\to\infty}\frac{1+2^3+.....n^3}{n^5}$ is equal to

1) $\frac{1}{30}$ 2) zero   3) $\frac{1}{4}$ 4) $\frac{1}{5}$

34. If $f(x)=\sqrt{\frac{x-\sin x}{x+\cos^2 x}}$, then $\lim_{x\to\infty}f(x)$ is

1) 0   2) ∞ 3) 1   4) -1

35. $\lim_{n\to\infty}\left\{\frac{1}{1-n^2}+\frac{2}{1-n^2}+....+\frac{n}{1-n^2}\right\}$ is equal to

1) 0   2) $-\frac{1}{2}$ 3) $\frac{1}{2}$ 4) 1

36. $\lim_{x\to-\infty}\left[\frac{x^4\sin\left(\frac{1}{x}\right)+x^2}{\left(1+|x|^3\right)}\right]=$ ______.

1) -1 2) 1   3) 2   4) 0

37. Let $f:R\to R$ be a positive increasing function with $\lim_{x\to\infty}\frac{f(3x)}{f(x)}=1$ then $\lim_{x\to\infty}\frac{f(2x)}{f(x)}=$

1) $\frac{2}{3}$ 2) $\frac{3}{2}$ 3) 3   4) 1

LEVEL-II (ADVANCED)

Single answer type questions

1. If $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$ then,

a) $a=1, b=4$ b) $a=1, b=-4$ c) $a=2, b=-3$ d) $a=2, b=3$

2. $\lim_{x \to 1} \frac{(1-x)(1-x^2).....(1-x^{2n})}{\{(1-x)(1-x^2)....(1-x^n)\}^2}, n \in N$ equals

a) ${}^{2n}p_n$ b) ${}^{2n}C_n$ c) $(2n)!$ d) n!

3. The value of $\lim_{x \to a} \sqrt{a^2-x^2} \cot \frac{\pi}{2} \sqrt{\frac{a-x}{a+x}}$ is

a) $\frac{2a}{\pi}$ b) $-\frac{2a}{\pi}$ c) $\frac{4a}{\pi}$ d) $-\frac{4a}{\pi}$

4. Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a}-1)x^2+(\sqrt{1+a}-1)x+(\sqrt[6]{1+a}-1)=0$ where $a>-1$. Then $\lim_{a \to 0^+} \alpha(a)$ and $\lim_{a \to 0^+} \beta(a)$ are

a) $-\frac{5}{2}$ and 1 b) $-\frac{1}{2}$ and -1

c) $-\frac{7}{2}$ and 2 d) $-\frac{5}{2}$ and $-\frac{7}{2}$

5. $\underset{x \to -1}{lt} \frac{1}{\sqrt{|x|-\{-x\}}}$ (where $\{x\}$ denotes the fractional part of x)

a) Does not exist b) equals1 c) equals 0 d) equals $\frac{1}{2}$

6. $\lim_{x \to 1} \frac{x \sin(x-[x])}{x-1}$,where [.]denotes the greatest integer function, is equal to

a) 0 b) -1 c) non–existent d) 2

7. The value of $\lim_{x \to 0} \frac{32}{x^8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}.\cos\frac{x^2}{4}\right)$

a) $\frac{1}{8}$ b) $\frac{3}{8}$ c) $\frac{7}{8}$ d) $\frac{9}{8}$

8. Let n be an odd integer. If $\sin n\theta = \sum_{r=0}^{n} b_r \sin^r \theta$, for every value of θ, then

a) $b_0=1, b_1=3$ b) $b_0=0, b_1=n$ c) $b_0=-1, b_1=n$ d) $b_0=0, b_1=n^2-3n+3$

9. Let m and n be two positive integers greater than 1. If $\lim_{\alpha \to 0} \frac{e^{\cos(\alpha^n)}-e}{\alpha^m} = -\frac{e}{2}$, then the value of $\frac{m}{n}$ is

a) 1 b) 2 c) 3 d) 4

10. If $\underset{x \to 0}{Lt}\, g(x)\left(\frac{e^{\frac{1}{x}}-e^{-\frac{1}{x}}}{e^{\frac{1}{x}}+e^{-\frac{1}{x}}}\right)$ exists, then $g(x) =$

a) x^3+3 b) x^2 c) x^2+3 d) x^2+4

11. $\lim_{x\to\infty} \frac{\log x^n - [x]}{[x]}, n \in N$, ([x] denotes the greatest integer less than or equal to x)

a) has the value – 1 b) has the value 0 c) has the value 1 d) does not exist

12. $\lim_{n\to\infty} \frac{1^p + 2^p + 3^p + ... + n^p}{n^{p+1}}$ is

a) $\frac{1}{P+1}$ b) $\frac{1}{1-p}$ c) $\frac{1}{p} - \frac{1}{p-1}$ d) $\frac{1}{p+2}$

13. $\lim_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2} + \frac{2}{n^2}\sec^2\frac{4}{n^2}......+\frac{1}{n}\sec^2 1\right]$ equals

a) $\frac{1}{2}\sec 1$ b) $\frac{1}{2}\text{cosec} 1$ c) $\tan 1$ d) $\frac{1}{2}\tan 1$

More than one correct answer type questions

14. If $f(x) = \frac{3x^2 + ax + a + 1}{x^2 + x - 2}$, the which of the following can be correct ?.

a) $\lim_{x\to 1} f(x)$ exists $\Rightarrow a = -2$ b) $\lim_{x\to -2} f(x)$ exists $\Rightarrow a = 13$

c) $\lim_{x\to 1} f(x) = 4/3$ d) $\lim_{x\to -2} f(x) = -1/3$

15. If $p = \underset{x\to 0^+}{Lt}\left[\frac{3\sin x}{x}\right], q = \underset{x\to 0^+}{Lt}\left[\frac{3x}{\sin x}\right], r = \underset{x\to 0^+}{Lt}\left[\frac{\tan x}{x}\right]$ and $S = \underset{x\to 0^+}{Lt}\left[\frac{3\tan x}{x}\right]$, where [.] denotes greatest integer function, then

a) $pqrs = 18$ b) $pqrs = 0$ c) $p + q + r + s = 10$ d) $pq + qr + rs + sp = 18$

16. Let $f(x) = \frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}.(1-\{x\})}$, where $\{x\}$ denotes the fractional part of x. Then

a) $\lim_{x\to 0+} f(x) = \infty$ b) $\lim_{x\to 0-} f(x) = \frac{\pi}{2\sqrt{2}}$ c) $\lim_{x\to 0+} f(x) = -\frac{\pi}{2}$ d) $\lim_{x\to 0-} f(x) = 0$

17. if $\lim_{x\to 1}(2 - x + a[x-1] + b[1+x])$ exists, then a and b can take the value (where[.] denotes the greatest integer function)

a) $a=1/3,\ b=1$ b) $a=1,\ b=-1$ c) $a = 9,\ b = -9$ d) $a = 2,\ b = 2/3$

18. Given a real-valued function f such that $f(x) = \begin{cases} \frac{\tan^2\{x\}}{(x^2 - [x]^2)}, & for\ x > 0 \\ 1 & for\ x = 0 \\ \sqrt{\{x\}\cot\{x\}}, & for\ x < 0 \end{cases}$ Where $[x]$ is the integral part and $\{x\}$ is the fractional part of x, then

a) $\underset{x\to 0^+}{Lt} f(x) = 1$ b) $\underset{x\to 0^-}{Lt} f(x) = \cot 1$ c) $\cot^{-1}\left(\underset{x\to 0^-}{Lt} f(x)\right)^2 = 1$ d) $\cot^{-1}\left(\underset{x\to 0^-}{Lt} f(x)\right)^2 + 1$

19. $\lim_{n\to\infty}\frac{-3n+(-1)^n}{4n-(-1)^n}$ is equals to

a) $-\frac{3}{4}$ b) 0 if n is even c) $-\frac{3}{4}$ *if n is odd* d) $\frac{1}{2}$

Matrix matching type questions

20. **Column - I** **Column - II**

A) If $\lim_{x\to\infty}(\sqrt{x^2-x-1}-ax-b)=0$, where $a>0$, then there exists at least one a and b for which point $(a, 2b)$ lies on the line p) $y=-3$

B) If $\lim_{x\to\infty}\frac{(1+a^3)+8e^{1/x}}{1+(1-b^3)e^{1/x}}=2$, then there exists at least one a and b for which point (a, b^3) lies on the line q) $3x-2y-5=0$

C) If $\lim_{x\to\infty}(\sqrt{x^4-x^2+1}-ax^2-b)=0$, then there exist at least one a and b for which point $(a, -2b)$ lies on the line r) $15x-2y-11=0$

D) If $\lim_{x\to -a}\frac{x^7+a^7}{x+a}=7$, where $a<0$, then there exists at leastone a for which point $(a, 2)$ lies on the line. s) $y=2$

21. **Column - I** **Column - II**

A) If $\lim_{x\to -1}\frac{\sqrt[3]{(7-x)}-2}{(x+1)}$, then 12L = p) -2

B) If $L=\lim_{x\to\pi/4}\frac{\tan^3 x-\tan x}{\cos\left(x+\frac{\pi}{4}\right)}$, then – L/4 q) 2

C) If $L=\lim_{x\to 1}\frac{(2x-3)(\sqrt{x}-1)}{2x^2+x-3}$, then 20L r) 1

D) If $L=\lim_{x\to\infty}\frac{\log x^n-[x]}{[x]}$, when $n\in N$, ([x] denotes greatest integer less than or equal to x) , then –2L s) -1

Integer answer type questions

22. Let $p(x)$ be a polynomial of degree 4 having extremum at x = 1, 2 and $\lim_{x\to 0}\left(1+\frac{p(x)}{x^2}\right)$ = 2. Then the value of $p(2)$ is ____

23. $\lim_{t \to 0}\left[Min(x^2+4x+6)\frac{\sin t}{t}\right]$ is equal to (where [.]denotes the G.I.F) =

24. The value of $\lim_{n \to \infty}\sum_{k=1}^{n}\frac{6^k}{(3^k-2^k)(3^{k+1}-2^{k+1})}$ is equal to

25. If $P=\lim_{n \to \infty}\left(1-\frac{1}{4}\right)\left(1-\frac{1}{9}\right)\left(1-\frac{1}{16}\right)\cdots\cdots\left(1-\frac{1}{n^2}\right)$; $n \geq 1$ then $\frac{1}{P}$ is equal to

26. $\underset{x \to \infty}{Lt}\frac{\sum_{r=1}^{10}(x+r)^{2010}}{(x^{1006}+1)(2x^{1004}+1)}=$................

EXERCISE-II

(Limit of the form 0^0, $1^\infty, \infty^0$, L.H. rule)

LEVEL-I (MAIN)

Single answer type questions

Limit of the form $0^0, 1^\infty, \infty^0$

1. $\underset{x \to 0}{Lt}\left\{(1+x)^{2/x}\right\}$ (where $\{x\}$ denotes fractional part of x) is equal to

1) e^2-5 2) e^2-8 3) e^2-6 4) e^2-7

2. The value of $\lim_{x \to 0}\left\{\sin^2\left(\frac{\pi}{2-ax}\right)\right\}^{\sec^2\frac{\pi}{2-bx}}$ is equal to

1) $e^{-a/b}$ 2) e^{-a^2/b^2} 3) $a^{2a/b}$ 4) $e^{4a/b}$

3. $\lim_{x \to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\operatorname{cosec} x}$ is equal to

1) e 2) $\frac{1}{e}$ 3) 1 4) e^2

4. The value of $\lim_{x \to 1}(2-x)^{\tan\left(\frac{\pi x}{2}\right)}$ is

1) $e^{-2/\pi}$ 2) $e^{1/\pi}$ 3) $e^{2/\pi}$ 4) $e^{-1/\pi}$

5. The value of $\lim_{m \to \infty}\left(\cos\frac{x}{m}\right)^m$ is

1) 1 2) e 3) e^{-1} 4) 2

6. $\lim_{n \to \infty}\left(\frac{n^2-n+1}{n^2-n-1}\right)^{n(n-1)}$ is equal to

1) e 2) e^2 3) e^{-1} 4) 1

7. $\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+....+n^x}{n}\right)^{1/x}$ is equal to

1) $(n!)^n$ 2) $(n!)^{\frac{1}{n}}$ 3) $n!$ 4) $\ln(n!)$

8. $\lim_{x\to\infty}\left(\frac{x^2+5x+3}{x^2+x+3}\right)^{1/x}$ is equal to

1) e^4 2) e^2 3) e^3 4) 1

9. If $\lim_{x\to\infty}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are

1) $a\in R, b\in R$ 2) $a=1, b\in R$ 3) $a\in R, b=2$ 4) $a=1, b=2$

10. The value of $\lim_{x\to 0}\left((\sin x)^{\frac{1}{x}}+\left(\frac{1}{x}\right)^{\sin x}\right)$, where $x>0$, is

1) 0 2) – 1 3) 1 4) 2

11. If $\lim_{x\to 0}[1+x\ln(1+b^2)]^{1/x}=2b\sin^2\theta, b>0$, smf $\theta\in(-\pi,\pi]$, then the value of θ is

1) $\pm\frac{\pi}{4}$ 2) $\pm\frac{\pi}{3}$ 3) $\pm\frac{\pi}{6}$ 4) $\pm\frac{\pi}{2}$

12. Let $f(x)$ be a twice differentiable function and $f''(0)=5$, then $\lim_{x\to 0}\frac{3f(x)-4f(3x)+f(9x)}{x^2}$ is equal to

1) 120 2) 119 3) 121 4) 118

13. $\lim_{x\to\infty}\frac{x(\log x)^3}{1+x+x^2}$ equals

1) 0 2) –1 3) 1 4) does not exist

14. If $\lim_{x\to 0}\frac{\log(3+x)-\log(3-x)}{x}=k$, the value of k is

1) 0 2) $-\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $-\frac{2}{3}$

15. Let $f(a)=g(a)=k$ and their n^{th} derivatives $f^n(a)$ and $g^n(a)$ exist and are not equal for some n. Further if $\lim_{x\to a}\frac{f(a)g(x)-f(a)-g(a)f(x)+g(a)}{g(x)-f(x)}=4$, then the value of k is

1) 4 2) 2 3) 1 4) 0

16. If $G(x)=-\sqrt{25-x^2}$, then $\lim_{x\to 1}\frac{G(x)-G(1)}{x-1}$ is

1) $\frac{1}{24}$ 2) $\frac{1}{5}$ 3) $-\sqrt{24}$ 4) $\frac{1}{2\sqrt{6}}$

17. $\lim_{h\to 0}\frac{(a+h)^2\sin(a+h)-a^2\sin a}{h}$.

1) $a^2\sin a+2a\sin a$ 2) $a^2\cos a+2a\sin a$ 3) $a^2\sin a+2a\cos a$ 4) $a^2\cos a+2a\cos a$

18. If $\lim_{x\to 0}\frac{\sin 3x+a\sin 2x}{x^3}=\lambda$ and λ is finite non zero real number, then $\lambda=$

1) $-\frac{1}{2}$ 2) $-\frac{3}{2}$ 3) $-\frac{5}{2}$ 4) $-\frac{7}{2}$

19. If $\lim_{x\to 0}\left(\frac{\sin 2x}{x^3}+a+\frac{b}{x^2}\right)=0$, then

1) $a=\frac{4}{3},b=2$ 2) $a=\frac{-4}{3},b=2$ 3) $a=\frac{4}{3},b=-2$ 4) $a=\frac{-4}{3},b=-2$

20. The value of $\lim_{x\to 0}\frac{\cos(\sin^2 x)-\cos(x^2)}{x^6}$ is

1) 0 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{3}{4}$

21. If $n=2$ then the value of $\lim_{x\to 0}\frac{e^x-1-x-\frac{x^2}{2!}-\dots\dots\dots-\frac{x^n}{n!}}{x^{n+1}}$

1) $\frac{1}{2}$ 2) $\frac{1}{6}$ 3) $\frac{1}{4}$ 4) $\frac{1}{8}$

22. $\lim_{x\to 0}\frac{e^{x^3}-1-x^3}{\sin^6(2x)}=$

1) $\frac{1}{128}$ 2) $\frac{2}{127}$ 3) $\frac{1}{126}$ 4) $\frac{1}{125}$

23. The value of $\lim_{x\to 0}\frac{1+\sin x-\cos x+\log(1-x)}{x^3}$ is

1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) 0 4) 2

24. If $\lim_{x\to 0}\left(\frac{x^n\sin^n x}{x^n-\sin^n x}\right)$ is non – zero finite, then n is equal to

1) 1 2) 2 3) 3 4) 4

25. If $[x]$ denotes greatest integer function then $\lim_{n\to\infty}\frac{[x]+[2x]+[3x]+---[nx]}{n^2}=$

1) $\frac{x}{2}$ 2) x 3) $\frac{x}{3}$ 4) $\frac{x}{4}$

26. $\lim_{x\to 0}\frac{\log(1+x+x^2)+\log(1-x+x^2)}{\sec x-\cos x}=$

1) –1 2) 1 3) 0 4) 2

27. If $\lim_{x\to a}[f(x)g(x)]$ exists, then both $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist.

1) true 2) false 3) true or false 4) neirther true or false

28. $\lim_{x\to 0}\frac{\log x^n-[x]}{[x]}, n\in N$, ([x] denotes greatest integer less than or equal to x)

1) has value -1 2) has value 0 3) has value 1 4) does not exist

LEVEL-II (ADVANCED)

Single answer type questions

1. $\lim_{x\to 1}\left[\operatorname{cosec}\frac{\pi x}{2}\right]^{1/(1-x)}$ (where [.] represents the greatest integer function) is equation to

a) 0 b) 1 c) ∞ d) does not exist

2. $\lim_{n\to\infty}\left\{\left(\frac{n}{n+1}\right)^{\alpha}+\sin\frac{1}{n}\right\}^{n}$ (where $\alpha\in Q$) is equal to

a) $e^{-\alpha}$ b) $-\alpha$ c) $e^{1-\alpha}$ d) $e^{1+\alpha}$

3. $\lim_{x\to\infty}\left[\left(\frac{e}{1-e}\right)\left(\frac{1}{e}-\frac{x}{1+x}\right)\right]^{x}$ is

a) $e^{(1-e)}$ b) $e^{\left(\frac{1-e}{e}\right)}$ c) $e^{\left(\frac{e}{1-e}\right)}$ d) $e^{\frac{1+e}{e}}$

4. The value of $Lim_{x\to\frac{\pi}{2}}\left(2^{\frac{1}{\cos^2 x}}+3^{\frac{1}{\cos^2 x}}+4^{\frac{1}{\cos^2 x}}+5^{\frac{1}{\cos^2 x}}+6^{\frac{1}{\cos^2 x}}\right)^{2\cos^2 x}$ is

a) 1 b) 6 c) 36 d) $\frac{1}{36}$

5. $Lt_{x\to 0}\left(\frac{(1+x)^{1/x}}{e}\right)^{1/x}=$

a) 1 b) e c) $\sqrt{e}$ d) $\frac{1}{\sqrt{e}}$

6. $Lt_{x\to 0}\log_{\cos\frac{x}{2}}\cos x=$

a) 1 b) 2 c) 3 d) 4

7. $\lim_{x\to\frac{\pi}{2}}\frac{1-(\sin x)^{\sin x}}{\cos^2 x}=$

a) 2 b) 1 c) 1/2 d) 1/4

8. If $f(x)=\dfrac{\cos x}{(1-\sin x)^{1/3}}$,then

a) $\lim\limits_{x\to\frac{\pi}{2}} f(x)=-\infty$ b) $\lim\limits_{x\to\frac{\pi}{2}+} f(x)=\infty$ c) $\lim\limits_{x\to\frac{\pi}{2}} f(x)=\infty$ d) $\lim\limits_{x\to\frac{\pi}{2}} f(x)=0$

9. The value of $\lim\limits_{x\to\infty}\dfrac{\left(2^{x^n}\right)^{\frac{1}{e^x}}-\left(3^{x^n}\right)^{\frac{1}{e^x}}}{x^n}$ (where $n\in N$) is

a) $\log n\left(\dfrac{2}{3}\right)$ b) 0 c) $n\log n\left(\dfrac{2}{3}\right)$ d) not defined

10. If f is a periodic function with period T & [.] denotes GIF, then,

$$\underset{n\to\infty}{Lt}\frac{\left[f(x+T)^{f(x+T)}\right]+\left[2^2 f(x+2T)^{f(x+2T)}\right]+\left[3^2 f(x+3T)^{f(x+3T)}\right]+\ldots\ldots+\left[n^2 f(x+nT)^{f(x+nT)}\right]}{n^3}$$

a) $\dfrac{(f(x))^{f(x)}}{2}$ b) $\dfrac{(f(x))^{f(x)}}{4}$ c) $\dfrac{2(f(x))^{f(x)}}{3}$ d) $\dfrac{(f(x))^{f(x)}}{3}$

More than one correct answer type questions

11. Which of the following statement(s) is/are correct

a) $\underset{x\to\frac{\pi}{2}}{Lt}\cot x.\ln(\sec x)=0$ b) $\underset{x\to\infty}{Lt}(\pi-2\tan^{-1}x)\ln x=0$

c) $\underset{x\to 0}{Lt}(\operatorname{cosec} x)^{\frac{1}{\ln x}}=1$ d) $\underset{x\to 0}{Lt}(\cot x)^{\frac{1}{\ln x}}=e$

12. In which of the following cases limit exists at the indicated points

a) $f(x)=\dfrac{\left[x+|x|\right]}{x}$ at $x=0$ where [x] denotes the greatest integer function

b) $f(x)=\dfrac{xe^{1/x}}{1+e^{1/x}}$ at $x=0$

c) $f(x)=(x-3)^{1/5}\,Sgn(x-3)$ at $x=3$, where sgn stands for signum function

d) $f(x)=\dfrac{\tan^{-1}|x|}{x}$ at $x=0$

Linked comprehension type questions

Passage - I :

$$L=\lim_{x\to 0}\frac{\sin x+ae^x+be^{-x}+c\ln(1+x)}{x^3}=$$

13. The value of L is

a) 1/2 b) −1/3 c) −1/6 d) 3

14. Equation $ax^2 + bx + c = 0$ has

a) real and equal roots
b) complex roots
c) unequal positive real roots
d) unequal roots

15. The solution set of $||x+c||-2a|<4b$ is

a) $[-2,2]$
b) $[0,2]$
c) $[-1,1]$
d) $[-2,1]$

Matrix matching type questions

16. Match the following $\lim_{x\to 0} f(x)$ where 'f' is given in column I with corresponding limit values in column-II.

Column - I	Column - II
A) $f(x)=\dfrac{\sin 3x}{\log(1+5x)}$	p) 4/5
B) $f(x)=\dfrac{e^{\sin 3x}-1}{\log(1+\tan 2x)}$	q) 3/5
C) $f(x)=\dfrac{\log(2-\cos 2x)}{(\log(\sin 3x+1))^2}$	r) 3/2
D) $f(x)=\dfrac{\log(1+\sin 4x)}{(e^{\sin 5x}-1)}$	s) 2/9

17. Match the following limit values.

Column - I	Column - II
A) $\lim_{x\to\infty}\left(\dfrac{x^2+x+1}{2x^2+3}\right)^x$	p) 1/2
B) $\lim_{x\to 0}\dfrac{1-\cos(1-\cos x)}{(e^{4x}-1)x^2.\sin x}$	q) -2
C) $\lim_{x\to 0}\dfrac{3}{64}\left(\dfrac{1}{x^2}-\cot^2 x\right)$	r) 1/32
D) $\lim_{x\to 0}\dfrac{[|x|]}{x}$	s) 0

Integer answer type questions

18. If $K=\lim_{x\to 0}\dfrac{f(x^2)-f(x)}{f(x)-f(0)}$ then $7 - K =$

19. $f(x)$ is the integral of $\dfrac{2\sin x-\sin 2x}{x^3}, x\neq 0$. Find $\lim_{x\to 0} f'(x)$ $\left[\text{where } f'(x)=\dfrac{df(x)}{dx}\right]$.

20. If $\underset{x\to 0}{Lt}\dfrac{\sin(\sin x)-\sin x}{ax^5+bx^3+c}=\dfrac{-1}{12}$ then b =

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL - I

1) 4	2) 3	3) 1	4) 4	5) 2	6) 4	7) 2	8) 2
9) 4	10) 2	11) 4	12) 3	13) 4	14) 4	15) 1	16) 4
17) 4	18) 3	19) 3	20) 4	21) 2	22) 2	23) 3	24) 1
25) 1	26) 3	27) 2	28) 1	29) 2	30) 1	31) 3	32) 2
33) 4	34) 3	35) 2	36) 1	37) 4			

LEVEL - II

1) b	2) b	3) c	4) b	5) a	6) c	7) a	8) b
9) b	10) b	11) a	12) a	13) d	14) abcd	15) ad	16) ab
17) bc	18) ab	19) ac	20) A-q; B-p; C-q; D- s				
21) A-s; B-r; C-p; D-q			22) 0	23) 1	24) 2	25) 2	26) 5

EXERCISE-II

LEVEL - I

1) 4	2) 2	3) 3	4) 3	5) 1	6) 2	7) 2	8) 4
9) 2	10) 3	11) 4	12) 1	13) 1	14) 3	15) 1	16) 4
17) 1	18) 3	19) 3	20) 3	21) 2	22) 1	23) 2	24) 2
25) 1	26) 2	27) 2	28) 4				

LEVEL - II

1) b	2) c	3) c	4) c	5) d	6) d	7) c	8) d
9) b	10) d	11) ab	12) abcd	13) b	14) b	15) c	
16) A-q; B-r; C-s; D-p			17) A-p; B-r; C-r; D-s			18) 8	19) 1
20) 2							